

MA3632 Lecture 10 – Model Evaluation and Selection

Throughout Lectures 5 to 9, cross-validation appeared repeatedly as a tool for choosing hyperparameters: the number of neighbours k , the regularisation strength λ , the tree depth d , the number of ensemble members B . In each case the procedure was the same – split the data, fit on part, evaluate on the rest, repeat – but the underlying statistical questions were left implicit. This lecture makes those questions explicit.

The central problem is that a model’s performance on the data used to fit it is an optimistic estimate of its performance on new data. Correcting for this optimism requires held-out data, but held-out data used for model selection is no longer genuinely held out. The proper use of training, validation, and test sets, the statistical properties of cross-validation estimators, the multiple-comparisons problem that arises when many models are compared, and the calibration of predicted probabilities are all aspects of the same underlying concern: how do we know how good a model really is?

1 The Train, Validate, Test Protocol

A common source of confusion is the conflation of two distinct uses of held-out data: *model selection* (choosing among candidate models or hyperparameter values) and *performance estimation* (reporting an honest estimate of the chosen model’s generalisation error). Using the same data for both invalidates the performance estimate.

Definition 1.1. The **train/validate/test protocol** divides the available labelled data $\mathcal{D}$ into three disjoint subsets.

- (i) The **training set** $\mathcal{D}_{\text{tr}}$ is used to fit model parameters (weights, split thresholds, etc.).
- (ii) The **validation set** $\mathcal{D}_{\text{val}}$ is used to select among candidate models or hyperparameter values. It is not used for fitting.
- (iii) The **test set** $\mathcal{D}_{\text{te}}$ is used exactly once, after all model selection decisions have been finalised, to report the estimated generalisation error of the chosen model.

The test set must be used exactly once. If the test error is inspected and then a different model is chosen, the test set has effectively become a second validation set and the reported error is no longer an unbiased estimate of generalisation error. This is sometimes called *test set contamination*.

Typical split proportions depend on dataset size. For large datasets ($n \geq 10^5$), a 70/15/15 or 80/10/10 split is common: there is enough data in each subset to give reliable estimates. For small datasets ($n \leq 10^3$), such splits leave too little data in each subset; cross-validation on the combined training and validation set is preferred, with a small test set held out at the very start of the analysis.

Remark 1.2. In competitions (e.g. Kaggle) and some academic benchmarks, the test set labels are not released to participants. The leaderboard score is computed on a public portion of the test set, and the final evaluation uses a private portion. This two-stage setup is designed to prevent participants from effectively selecting models using test set feedback, but in practice the public leaderboard still induces selection bias when many submissions are made.

2 Cross-Validation Revisited

When data are limited, a fixed validation set wastes too much data for evaluation. Cross-validation addresses this by rotating the validation role across different subsets of $\mathcal{D}_{\text{tr}}$.

Definition 2.1. In *k -fold cross-validation*, the training set $\mathcal{D}_{\text{tr}}$ is partitioned into k folds $F_1, \dots, F_k$ of approximately equal size. For each fold $j \in \{1, \dots, k\}$, a model is trained on $\mathcal{D}_{\text{tr}} \setminus F_j$ and evaluated on F_j . The cross-validation estimate of generalisation error is

$$\widehat{\text{CV}}_k = \frac{1}{k} \sum_{j=1}^k \hat{R}_j,$$

where $\hat{R}_j$ is the empirical risk on fold j .

2.1 What does cross-validation estimate?

A precise understanding of what $\widehat{\text{CV}}_k$ estimates is important for correct interpretation. Each fold j trains a model on $(1 - 1/k)$ of the training data. $\widehat{\text{CV}}_k$ can be interpreted as estimating the expected generalisation performance of the *learning procedure* when trained on approximately $(1 - 1/k)n$ observations, rather than the performance of the specific final fit that uses all n observations.

Proposition 2.2. *For a learning algorithm whose generalisation error is a decreasing function of training set size (as is typical), $\widehat{\text{CV}}_k$ overestimates the generalisation error of the final model trained on all n observations. The training-size component of this bias generally decreases as k increases, since $(1 - 1/k)n$ approaches n ; whether it is negligible in practice for $k = 10$ depends on the learning algorithm, the dataset, and the shape of the learning curve.*

The special case $k = n$ is leave-one-out cross-validation (LOOCV). Each model is trained on $n - 1$ observations, so the training-size component of the CV bias is smallest among ordinary k -fold choices. However, LOOCV can have high variance: heavy overlap among the n training sets (each pair shares $n - 2$ of their $n - 1$ points) can induce substantial correlation among the n error estimates, which in turn can inflate the variance of their average relative to the idealised independent case; see Proposition 2.3. For most purposes, $k = 5$ or $k = 10$ provides a better bias-variance trade-off for the CV estimator itself.

Proposition 2.3. *Suppose the LOOCV errors $e_1, \dots, e_n$ are identically distributed with variance σ_e^2 and have a common pairwise covariance $\text{Cov}(e_i, e_j) = c$ for all $i \neq j$ (an idealisation that holds exactly only under suitable exchangeability conditions on the data and learning algorithm). Then the variance of $\widehat{\text{CV}}_n = n^{-1} \sum_{i=1}^n e_i$ is*

$$\text{Var}(\widehat{\text{CV}}_n) = \frac{1}{n^2} [n \sigma_e^2 + n(n - 1) c] = \frac{\sigma_e^2}{n} + \frac{n - 1}{n} c.$$

Because the n leave-one-out training sets overlap heavily, the LOOCV errors are typically not independent, and this heavy overlap can induce a substantial common covariance c , inflating the variance of the LOOCV estimator above the σ_e^2/n that would hold under independence. This is a heuristic justification, not a general theorem that $c > 0$ in every setting, but it is the usual reason moderate k (such as $k = 5$ or $k = 10$) is often preferred over LOOCV in practice.

2.2 Stratified cross-validation

For classification problems with imbalanced classes, random partitioning into folds may produce folds with very different class distributions, making the CV estimate noisy. **Stratified k -fold cross-validation** ensures that each fold contains approximately the same proportion of each class as the full training set. This is the default in scikit-learn's **StratifiedKFold** and is standard practice for classification.

2.3 Repeated cross-validation

The randomness in fold assignment introduces additional variance into $\widehat{CV}_k$. **Repeated k -fold cross-validation** runs k -fold CV multiple times (each with a different random partition) and averages the results. With r repetitions of k -fold CV, the number of model fits is rk . Averaging across repetitions reduces the variability due to the particular random fold assignment, although the reduction is generally less than the $1/r$ factor that would apply to independent repetitions, because all repetitions use the same underlying observations and are therefore correlated with one another. Repeated 5-fold CV with $r = 10$ (50 fits total) is often recommended as a good trade-off between computational cost and estimator stability.

3 The Multiple Comparisons Problem

A typical model selection procedure evaluates many candidate models and selects the one with the lowest CV error. This selection step introduces a bias: the chosen model is likely to have a lower CV error than its true generalisation error, simply because it was chosen as the minimum over a collection of noisy estimates.

Proposition 3.1. *Let $\hat{R}_1, \dots, \hat{R}_M$ be the CV errors of M candidate models, each an unbiased estimate of the corresponding true generalisation error $R_1, \dots, R_M$. Then $\mathbb{E}[\min_m \hat{R}_m] \leq \min_m R_m$: the minimum observed CV error underestimates the true error of the best model.*

Proof. $\min_m \hat{R}_m \leq \hat{R}_{m^*}$ where $m^* = \arg \min_m R_m$ is the truly best model. Taking expectations, $\mathbb{E}[\min_m \hat{R}_m] \leq \mathbb{E}[\hat{R}_{m^*}] = R_{m^*} = \min_m R_m$. $\square$

This is a form of the *winner's curse*: the model that wins the selection competition tends to have been lucky, not just good. In typical settings, this selection optimism tends to become more severe as more candidate models are searched and as their performance estimates become noisier, although this is not a consequence of Proposition 3.1 alone, which places no restriction on how the gap scales with M or with the noise level. The practical consequence is that the CV error of the selected model is an overly optimistic estimate of its true generalisation error.

3.1 The one-standard-error rule

One practical mitigation, already introduced in Lecture 5, is the **one-standard-error rule**: rather than selecting the model with the lowest CV error, select the simplest model whose CV error lies within one standard error of the minimum, where the relevant threshold is the minimum estimated CV error plus the estimated standard error of *that* minimum-error model (rather than, say, the standard error of the candidate model being compared against it). This rule acknowledges the uncertainty in the estimated CV error and prefers parsimony when performance is approximately equal.

3.2 Nested cross-validation

The principled solution to the multiple comparisons problem is **nested cross-validation** (also called *double cross-validation*).

Definition 3.2. Nested $k_{\text{out}} \times k_{\text{in}}$ **cross-validation** proceeds as follows.

- (i) Split $\mathcal{D}_{\text{tr}}$ into k_{out} outer folds $F_1^{\text{out}}, \dots, F_{k_{\text{out}}}^{\text{out}}$.
- (ii) For each outer fold j :
 - (a) Let the outer training set be $\mathcal{D}_j^{\text{out}} = \mathcal{D}_{\text{tr}} \setminus F_j^{\text{out}}$.
 - (b) Run k_{in} -fold CV on $\mathcal{D}_j^{\text{out}}$ to select the best hyperparameter $\hat{\lambda}_j$.
 - (c) Retrain the model with $\hat{\lambda}_j$ on all of $\mathcal{D}_j^{\text{out}}$ and evaluate it on F_j^{out} . Record the error $\hat{R}_j^{\text{out}}$.
- (iii) The nested CV estimate of generalisation error is $\widehat{\text{CV}}_{\text{nested}} = k_{\text{out}}^{-1} \sum_{j=1}^{k_{\text{out}}} \hat{R}_j^{\text{out}}$.

Nested CV separates model selection (inner loop) from performance estimation (outer loop). The outer error estimates are not contaminated by the model selection step because each outer fold was not used during the inner CV. The total number of model fits is $k_{\text{out}} \times k_{\text{in}} \times |\Lambda|$ for the inner grid search across all outer folds, plus a further k_{out} refits – one per outer fold, using that fold’s selected hyperparameter, as in step (ii)(c) above – before evaluation on the held-out outer fold; any final refit on the entire dataset after nested CV has completed (to produce the model actually deployed) is additional again. Typical choices are $k_{\text{out}} = 5$ and $k_{\text{in}} = 5$, giving $25|\Lambda| + 5$ fits for the nested procedure itself, which can be large when $|\Lambda|$ is large.

Remark 3.3. A common point of confusion after nested CV is how to obtain a final model to deploy. Taking the hyperparameter $\hat{\lambda}_j$ that was selected most often across the outer folds and retraining on all of $\mathcal{D}_{\text{tr}}$ with that value is sometimes done, but the cleaner and more standard workflow is to treat nested CV purely as an estimate of the generalisation error of the selection *procedure*, and separately rerun that same procedure – ordinary k_{in} -fold CV over Λ – on the entire available training set $\mathcal{D}_{\text{tr}}$ to choose the final hyperparameter, before fitting the final model on all of $\mathcal{D}_{\text{tr}}$ with that choice. Either way, the nested CV error estimate refers to the procedure (select and fit), not to any one specific model, and the final model’s true generalisation error may differ from the nested CV estimate, though for stable selection procedures the difference is typically small.

4 Statistical Comparison of Two Classifiers

When two models have been evaluated on the same test set, one naturally asks whether the difference in performance is statistically significant or merely due to chance. Naive approaches – comparing means of CV error estimates, or reporting p -values from two-sample t -tests on CV folds – are unreliable because the CV errors from the same folds are dependent.

4.1 McNemar’s test

For two classifiers evaluated on the same test set of n observations, the outcome of each classifier on each observation is either correct (C) or incorrect (I). The 2×2 table of joint outcomes is:

	Classifier B correct	Classifier B incorrect
Classifier A correct	n_{CC}	n_{CI}
Classifier A incorrect	n_{IC}	n_{II}

The observations in the n_{CC} and n_{II} cells do not distinguish the two classifiers. Only the n_{CI} observations (A correct, B incorrect) and n_{IC} observations (A incorrect, B correct) are informative.

Definition 4.1. McNemar’s test tests the null hypothesis $H_0 : P(\text{A correct, B incorrect}) = P(\text{A incorrect, B correct})$, i.e. that neither classifier is systematically better. Under H_0 , the counts n_{CI} and n_{IC} are exchangeable, and the test statistic

$$\chi^2 = \frac{(n_{CI} - n_{IC})^2}{n_{CI} + n_{IC}}$$

follows a chi-squared distribution with one degree of freedom (for large $n_{CI} + n_{IC}$). A continuity-corrected version replaces the numerator with $(|n_{CI} - n_{IC}| - 1)^2$.

McNemar’s test is appropriate for paired comparisons on a *fixed* test set. It accounts for the dependence between the two classifiers’ errors (since both are evaluated on the same observations) and is more powerful than comparing error rates using an unpaired test.

4.2 The 5×2 cross-validation test

McNemar’s test requires a fixed test set. When performance is estimated by CV, the 5×2 CV test of Dietterich (1998) is a statistically sound alternative.

Definition 4.2. The 5×2 cross-validation test compares two algorithms A and B as follows.

- (i) Repeat five times (for $i = 1, \dots, 5$): randomly split $\mathcal{D}$ into two equal halves $S_1^{(i)}$ and $S_2^{(i)}$. For each half, train A on that half and evaluate on the other, and similarly for B . Let $p_j^{(i)} = \hat{R}_{A,j}^{(i)} - \hat{R}_{B,j}^{(i)}$ be the difference in error rates for split $j \in \{1, 2\}$ at repetition i .
- (ii) Compute $\bar{p}^{(i)} = (p_1^{(i)} + p_2^{(i)})/2$ and $s_i^2 = (p_1^{(i)} - \bar{p}^{(i)})^2 + (p_2^{(i)} - \bar{p}^{(i)})^2$ for each i .
- (iii) The test statistic is

$$t = \frac{p_1^{(1)}}{\sqrt{\frac{1}{5} \sum_{i=1}^5 s_i^2}}.$$

- (iv) Under H_0 : no difference in generalisation error, t approximately follows a t -distribution with 5 degrees of freedom.

The 5×2 CV test is preferred over naive k -fold CV t -tests because its variance estimate is constructed from the paired differences obtained from the two complementary train/test directions within each of five independent 50/50 replications, specifically to address the dependence that invalidates treating ordinary k -fold CV results as independent observations.

5 Calibration of Predicted Probabilities

A classifier that outputs class probabilities may be highly accurate in terms of 0–1 loss while having poorly calibrated probabilities. Calibration refers to the agreement between predicted probabilities and observed frequencies: if a model assigns probability 0.7 to the positive class for a group of observations, then approximately 70% of those observations should actually be positive.

Definition 5.1. A probabilistic classifier is **perfectly calibrated** if

$$\mathbb{E}[Y \mid \hat{p}(\mathbf{X})] = \hat{p}(\mathbf{X}) \quad \text{almost surely,}$$

where $\hat{p}(\mathbf{x})$ is the predicted probability of class 1. (Conditioning directly on the event $\{\hat{p}(\mathbf{X}) = p\}$ for every individual $p \in [0, 1]$, as in the more elementary-looking statement $P(Y = 1 \mid \hat{p}(\mathbf{X}) = p) = p$, is not formally well defined when $\hat{p}(\mathbf{X})$ is continuously distributed, since that event can then have probability zero for every p ; the conditional-expectation form above is the precise version of the same idea.) Informally: among observations assigned a predicted probability of approximately p , approximately a proportion p of them should belong to the positive class.

Perfect calibration is a population-level condition that cannot be verified from finite data, but it can be assessed empirically using a **reliability diagram** (or *calibration curve*).

Definition 5.2. A **reliability diagram** is constructed as follows. Bin the predicted probabilities $\hat{p}(\mathbf{x}_i)$ into B equal-width intervals $[0, 1/B), [1/B, 2/B), \dots, [(B-1)/B, 1]$. For each bin b , compute the mean predicted probability $\bar{p}_b$ and the observed fraction of positives $\bar{y}_b$. Plot $\bar{y}_b$ against $\bar{p}_b$. A perfectly calibrated classifier produces a diagonal line; systematic deviations indicate miscalibration.

Two common patterns of miscalibration arise in practice. An *overconfident* classifier produces predicted probabilities that are too extreme relative to the observed outcome frequencies: probabilities near 0 and 1 are assigned more often, and with more extremity, than the actual class frequencies at those predicted values justify, so the reliability curve lies above the diagonal at low predicted probabilities and below the diagonal at high predicted probabilities. Decision trees and naive Bayes classifiers are often prone to this pattern. An *underconfident* classifier produces predicted probabilities that are too close to 0.5; SVM decision-function scores passed through a fixed transformation are a common example. Different classifiers can exhibit different characteristic calibration patterns, and averaging procedures such as bagging can behave either way depending on the data: averaging over many trees, each with somewhat extreme individual leaf predictions, often compresses the ensemble's probabilities toward the centre, but the net effect for Random Forests and other bagged ensembles is data- and model-dependent and should always be assessed empirically with a reliability diagram rather than assumed in advance.

5.1 The Brier score

A single-number summary of overall probabilistic prediction quality, reflecting both calibration and resolution, is the Brier score.

Definition 5.3. The **Brier score** for binary classification is

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}(\mathbf{x}_i) - y_i)^2.$$

A Brier score of 0 indicates perfect prediction; a score of 1 indicates worst possible prediction. A classifier that always predicts $\hat{p} = \bar{y}$ (the base rate) achieves $\text{BS} = \bar{y}(1 - \bar{y})$.

The Brier score can be decomposed (the Murphy decomposition) into reliability (calibration), resolution, and uncertainty components, making it a more informative summary than accuracy or AUC alone. A model can have high AUC (good ranking) but high Brier score (poor probability estimates), or conversely.

5.2 Post-hoc calibration methods

When a model’s probabilities are miscalibrated, they can often be corrected without refitting the full model. Two standard methods are available.

Platt scaling fits a logistic regression model with the raw model scores (e.g. an SVM decision function, or another continuous classifier score) as the single input feature and the true labels as the target. The logistic regression is fitted on a held-out calibration set (not the training set used to fit the original model). Platt scaling is appropriate when a logistic mapping from the model’s score to the event probability provides an adequate calibration model, as is often the case for SVMs.

Isotonic regression fits a piecewise-constant non-decreasing function to the empirical relationship between model scores and observed frequencies. It makes no parametric assumption about the shape of the miscalibration and is more flexible than Platt scaling, but requires more calibration data to estimate reliably. It is a natural choice when the observed miscalibration pattern is not well approximated by a single sigmoid, which can occur for Random Forests, boosted models, or other ensembles; the appropriate calibration method for any specific fitted model should nonetheless be confirmed empirically, for example by comparing calibration performance on held-out data, rather than assumed from the model class alone.

Both methods are implemented in scikit-learn’s `CalibratedClassifierCV`, which handles the train/calibration split automatically using cross-validation.

6 Addendum: Shapley Additive Explanations (SHAP)

The evaluation tools of Sections 1–5 answer how good a model is. They say nothing about why a model produced a particular prediction. This addendum introduces a single principled answer to that question, based on a solution concept from cooperative game theory: the Shapley value.

6.1 Feature attribution and the Shapley value

Fix a trained model f and an instance $\mathbf{x} = (x_1, \dots, x_d)$ whose prediction is to be explained. Write $N = \{1, \dots, d\}$ for the index set of features. The idea is to treat the d features as d players in a cooperative game whose payoff is the model’s prediction, and to ask how much each player (feature) contributed to that payoff.

Definition 6.1. The **value function** $v_{\mathbf{x}} : 2^N \rightarrow \mathbb{R}$ associated with model f and instance $\mathbf{x}$ assigns to each coalition $S \subseteq N$ the expected model output when the features in S are fixed at their values in $\mathbf{x}$ and the remaining features are averaged over the data distribution:

$$v_{\mathbf{x}}(S) = \mathbb{E}[f(\mathbf{X}) \mid \mathbf{X}_S = \mathbf{x}_S].$$

In particular $v_{\mathbf{x}}(\emptyset) = \mathbb{E}[f(\mathbf{X})]$ (the average prediction over the data, with no features fixed) and $v_{\mathbf{x}}(N) = f(\mathbf{x})$ (the actual prediction, with every feature fixed).

Remark 6.2. The conditional expectation $\mathbb{E}[f(\mathbf{X}) \mid \mathbf{X}_S = \mathbf{x}_S]$ above is one possible definition of the Shapley game underlying $v_{\mathbf{x}}$, and is the definition Lundberg and Lee (2017) themselves use. In practice, however, computing this conditional expectation exactly is often intractable, and implementations approximate it under simplifying assumptions about the missing features. For example, `KernelExplainer` with a supplied background dataset masks a missing feature by replacing it with values drawn from that background data – effectively using a *marginal* or *background-based* approximation to the conditional game rather than the conditional expectation itself. `TreeSHAP` similarly supports more than one feature-dependence convention (for example, *interventional* and

tree_path_dependent modes). These choices can give substantially different attributions from one another, and from the conditional formulation defined here, when features are strongly correlated. This lecture adopts the conditional-expectation formulation for concreteness and because it has a clean probabilistic motivation, but practitioners should check which convention a given software implementation actually uses.

Definition 6.3. For feature $i \in N$, the **Shapley value** of feature i for instance $\mathbf{x}$ is

$$\phi_i(v_{\mathbf{x}}) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(d - |S| - 1)!}{d!} [v_{\mathbf{x}}(S \cup \{i\}) - v_{\mathbf{x}}(S)].$$

The weight $|S|!(d - |S| - 1)!/d!$ is exactly the probability that, in a uniformly random ordering of the d features, the features in S appear before feature i and every other feature appears after it. The Shapley value is therefore the marginal contribution of feature i to the prediction, averaged over every possible order in which the features could be revealed.

6.2 Computing Shapley values in practice

Definition 6.3 sums over all 2^{d-1} subsets $S \subseteq N \setminus \{i\}$ for each of the d features, an exponential computation that is infeasible beyond $d \approx 15$ features directly. Two practical algorithms make Shapley-value attribution usable at the scale of real datasets.

Definition 6.4. KernelSHAP (Lundberg and Lee, 2017) approximates $\phi_1(v_{\mathbf{x}}), \dots, \phi_d(v_{\mathbf{x}})$ for an arbitrary model f by sampling coalitions S according to the **SHAP kernel** weighting

$$\pi(S) = \frac{d - 1}{\binom{d}{|S|} |S| (d - |S|)}, \quad 0 < |S| < d,$$

evaluating $v_{\mathbf{x}}(S)$ for each sampled coalition, and solving a weighted least-squares regression of these values on the binary coalition indicators subject to the efficiency constraint (Proposition 6.5 below). The regression coefficients recovered are the Shapley values. **TreeSHAP** (Lundberg et al., 2018) instead exploits the recursive structure of a decision tree to compute Shapley values *exactly*, for a particular choice of feature-dependence convention, for tree-based models (decision trees, Random Forests, gradient boosting) by tracking, for every root-to-leaf path, which fraction of training instances would reach that leaf under each possible coalition – at a cost polynomial in the number of trees, leaves, and tree depth, rather than exponential in d .

6.3 Properties: efficiency, and a comparison with MDI importance

Proposition 6.5 (Efficiency). *For any instance $\mathbf{x}$,*

$$\sum_{i=1}^d \phi_i(v_{\mathbf{x}}) = v_{\mathbf{x}}(N) - v_{\mathbf{x}}(\emptyset) = f(\mathbf{x}) - \mathbb{E}[f(\mathbf{X})].$$

Proof. Fix a uniformly random ordering σ of N and let $S_i(\sigma)$ be the set of features preceding i in σ . Summing the telescoping differences $v_{\mathbf{x}}(S_i(\sigma) \cup \{i\}) - v_{\mathbf{x}}(S_i(\sigma))$ over all i in the order given by σ collapses to $v_{\mathbf{x}}(N) - v_{\mathbf{x}}(\emptyset)$ for every fixed σ , since each feature is added exactly once as σ is processed from the empty coalition to N . Averaging this identity over all $d!$ orderings σ gives, on the left-hand side, exactly $\sum_i \phi_i(v_{\mathbf{x}})$ by Definition 6.3, and the right-hand side is unchanged by the averaging. $\square$

Proposition 6.5 is the reason SHAP values are called an *additive* feature attribution: the d attributions and the baseline $\mathbb{E}[f(\mathbf{X})]$ add up exactly to the prediction being explained, so $\phi_i(v_{\mathbf{x}})$ can be read directly as “feature i pushed this prediction up (or down) by this much, relative to the average prediction.”

Remark 6.6 (Consistency, and comparison with MDI importance). A **global** importance score for feature i can be obtained from Shapley values by averaging their magnitude over the dataset, $\frac{1}{n} \sum_{k=1}^n |\phi_i(v_{\mathbf{x}_k})|$, giving a ranking directly comparable to the MDI (Gini) importance of Lecture 9. The Shapley value itself has a guarantee that MDI importance lacks: it is **consistent**, meaning that if a model changes so that a feature’s marginal contribution weakly increases for every possible coalition S (for a fixed instance $\mathbf{x}$), that feature’s Shapley value for that instance cannot decrease. MDI importance does not satisfy the analogous property; it can rank a feature as more important in a model where the feature’s true influence on the prediction has, if anything, gone down, an artefact of how impurity reduction is attributed to splits rather than to the feature’s actual effect on predictions. This consistency guarantee is stated for the local, per-instance attribution under a specified model change; extending it to claim that the averaged, dataset-level importance ranking constructed above inherits exactly the same guarantee under arbitrary model changes would require additional argument beyond what is established here. MDI and mean absolute Shapley importance measure related but distinct quantities and can differ substantially in practice, particularly with correlated predictors, given MDI’s known high-cardinality bias (Lecture 8).

Theorem 6.7 (Shapley, 1953). *The Shapley value $\phi_i(v_{\mathbf{x}})$ is the unique attribution rule satisfying all of the following: (i) efficiency (Proposition 6.5); (ii) symmetry, that features contributing identical marginal values to every coalition receive equal attributions; (iii) the null player property, that a feature whose addition never changes the value of any coalition receives attribution zero; and (iv) additivity, that attributions for a sum of two value functions equal the sum of the attributions for each value function separately.*

Theorem 6.7 is the axiomatic justification for using the Shapley value specifically, rather than one of the many other plausible ways of splitting credit among features: it is the only rule with all four properties simultaneously. We omit the proof, which is a counting argument over the 2^d coalitions; see Shapley (1953) or Lundberg and Lee (2017, Theorem 1) for the attribution-specific statement.

7 Exercises

Exercise 7.1. A researcher uses 10-fold cross-validation to compare 50 machine learning models on a dataset with $n = 500$ observations. They select the model with the lowest CV error and report this CV error as the estimated generalisation error of their chosen model.

Identify two statistical problems with this procedure. For each problem, state what the consequence is and describe how it should be corrected. What would a correct procedure look like if the researcher’s goal is (a) to select the best model and report an honest estimate of its generalisation error, and (b) to select a model that is likely to perform well in practice even if the error estimate is slightly conservative?

Solution. **Problem 1: Multiple comparisons / selection bias.** Comparing 50 models and selecting the minimum CV error exploits randomness in the CV procedure. Even if all 50 models have the same true generalisation error, the minimum of 50 noisy estimates will be lower than any individual estimate (Proposition 3.1). The reported CV error therefore underestimates the true error of the chosen model. In typical settings, the selection bias tends to become more severe as

the number of models compared increases. Correction: use nested CV to separate selection from evaluation, so the outer error estimate is not contaminated by the inner selection step.

Problem 2: Lack of independent performance evaluation (reuse of validation information). There is no independent test set anywhere in this procedure: the same CV error that served as the validation criterion for selecting among the 50 models is then reported as the performance estimate. Having already been used to make the selection decision, this CV error is no longer an honest estimate of the selected model's generalisation error. Correction: either reserve a genuine test set that plays no role in selection and report performance on it (used exactly once), or, if data are too scarce to spare a dedicated test set, use nested CV so that the outer-loop estimate of the whole selection procedure is never contaminated by the inner selection step.

Correct procedure for goal (a) – two acceptable alternatives: (i) hold out a genuine test set at the outset, use ordinary CV on the remaining training data for model selection, then report performance on the untouched test set; or (ii) if the dataset is too small to spare a separate test set, use nested CV on the whole dataset to estimate the generalisation error of the selection procedure itself, reporting the nested CV error as the performance estimate. These two options serve the same purpose and should not normally both be applied to the same data.

Correct procedure for goal (b): use the one-standard-error rule within a standard CV procedure to select the simplest model whose CV error is within one SE of the minimum; this guards against overfitting the model selection step even without formal nested CV. The CV error of the selected model will still be slightly optimistic, but the effect is mitigated by preferring simpler models. $\square$

Exercise 7.2. Two classifiers A and B are evaluated on a test set of 200 observations. The results are summarised in the following table:

	B correct	B incorrect
A correct	140	30
A incorrect	10	20

Compute the error rates of A and B. Apply McNemar's test (without continuity correction) and state the test statistic and its approximate p -value. At the 5% significance level, is there a statistically significant difference between the two classifiers? What would the continuity-corrected statistic be?

Solution. Error rates: A makes $10 + 20 = 30$ errors, so $\hat{R}_A = 30/200 = 0.15$. B makes $30 + 20 = 50$ errors, so $\hat{R}_B = 50/200 = 0.25$.

From the table: $n_{CI} = 30$ (A correct, B incorrect), $n_{IC} = 10$ (A incorrect, B correct).

McNemar's statistic (uncorrected):

$$\chi^2 = \frac{(30 - 10)^2}{30 + 10} = \frac{400}{40} = 10.0.$$

Under H_0 , $\chi^2 \sim \chi_1^2$. The critical value at the 5% level is 3.84 and the p -value is $P(\chi_1^2 \geq 10.0) \approx 0.0016$. Since $10.0 > 3.84$, we reject H_0 at the 5% level and conclude that classifier A is significantly better than B.

Continuity-corrected statistic:

$$\chi_c^2 = \frac{(|30 - 10| - 1)^2}{30 + 10} = \frac{361}{40} = 9.025,$$

which still exceeds 3.84, so the conclusion is unchanged. $\square$

Exercise 7.3. A logistic regression model and a Random Forest are each fitted on the same training set and evaluated on the same test set of 500 observations. The predicted probabilities $\hat{p}_i$ for the positive class are available for both models.

The logistic regression produces a well-calibrated reliability diagram (close to the diagonal). In this particular case, the Random Forest produces a reliability diagram that is S-shaped: predicted probabilities near 0 and 1 are too extreme (the curve lies above the diagonal for low predicted probabilities and below the diagonal for high predicted probabilities).

Identify the type of miscalibration exhibited by the Random Forest in this example. This is not a universal property of Random Forests; briefly explain why averaging over many trees could just as easily compress predicted probabilities toward 0.5 instead. State which post-hoc calibration method (Platt scaling or isotonic regression) would be a reasonable starting point for correcting the miscalibration observed here, and explain what should ultimately determine the choice between the two methods. Finally, explain why calibration matters in a medical screening application where the model's predicted probabilities are used directly to triage patients.

Solution. The S-shaped reliability diagram indicates *overconfidence* in this instance: predicted probabilities near 0 and 1 appear more frequently than the corresponding observed frequencies justify. Since the curve lies above the diagonal for low predicted probabilities and below the diagonal for high predicted probabilities, when the model predicts $\hat{p} = 0.1$ the true positive rate among those observations is higher than 10%, and when it predicts $\hat{p} = 0.9$ the true positive rate is lower than 90%. The predicted probabilities are too extreme in both directions.

This is not a general property of Random Forests. Individual leaf predictions, based on the class proportions among the small number of training points reaching that leaf, can indeed be extreme. But averaging over many trees, each built on a different bootstrap resample with different splits, more commonly pulls the ensemble's aggregated probability toward the centre, producing under-confidence rather than overconfidence. Whether a fitted Random Forest turns out to be over- or under-confident depends on the dataset, the tree depth, the number of trees, and other modelling choices, and should be assessed empirically with a reliability diagram rather than assumed in advance, as in Section 5.

Both Platt scaling and isotonic regression are defensible candidates here, and neither is clearly ruled in or out by the S-shape alone: a logistic (sigmoid) recalibration map can itself correct an S-shaped distortion in the reliability diagram, so the visual shape of the miscalibration does not by itself determine which method is preferable. Isotonic regression offers greater flexibility – it makes no parametric assumption about the shape of the miscalibration – when sufficient calibration data are available to estimate it reliably; Platt scaling provides a lower-variance parametric calibration when a logistic mapping from the raw score to the event probability is adequate, and can be preferable with a limited calibration set even if the true miscalibration is not exactly sigmoidal. The preferred method should ultimately be selected by comparing calibration performance (for example, the Brier score) on held-out calibration data, using cross-validation if calibration data are limited.

In medical screening, the predicted probability directly determines clinical action: for example, a patient with $\hat{p} > 0.6$ might be referred for further testing. If the model is overconfident and predicts $\hat{p} = 0.8$ for patients whose true risk is 0.6, the threshold calibration is wrong and some patients may be incorrectly triaged. Worse, if a low-risk patient has a predicted probability of 0.15 when the true risk is 0.25, they may be incorrectly assigned to a low-priority group. Accurate calibration ensures that the numerical probability output can be used directly for clinical decision-making, not just for ranking patients. $\square$

Exercise 7.4. Define the Brier score for binary classification. Show that a classifier that always predicts $\hat{p}(\mathbf{x}) = \bar{y}$ (the base rate) achieves $BS = \bar{y}(1 - \bar{y})$. Show also that any classifier whose

empirical Brier score is no larger than that of the base-rate predictor satisfies $BS \leq \bar{y}(1 - \bar{y})$. Hence argue that $\bar{y}(1 - \bar{y})$ is a natural benchmark for the Brier score, analogous to $R^2 = 0$ for regression.

Solution. The Brier score is $BS = n^{-1} \sum_{i=1}^n (\hat{p}_i - y_i)^2$.

For $\hat{p}_i = \bar{y}$ for all i :

$$BS = \frac{1}{n} \sum_{i=1}^n (\bar{y} - y_i)^2 = \frac{1}{n} [n_1(\bar{y} - 1)^2 + n_0(\bar{y} - 0)^2],$$

where $n_1 = \sum y_i$ and $n_0 = n - n_1$, so $\bar{y} = n_1/n$. Then

$$BS = \bar{y}(\bar{y} - 1)^2 + (1 - \bar{y})\bar{y}^2.$$

Expanding: $\bar{y}(\bar{y}^2 - 2\bar{y} + 1) + (1 - \bar{y})\bar{y}^2 = \bar{y}^3 - 2\bar{y}^2 + \bar{y} + \bar{y}^2 - \bar{y}^3 = \bar{y} - \bar{y}^2 = \bar{y}(1 - \bar{y})$. This is exactly the empirical identity $n^{-1} \sum_i (\bar{y} - y_i)^2 = \bar{y}(1 - \bar{y})$ for the sample prevalence $\bar{y}$, with no distributional assumption required.

For the second part: this is now immediate. If a classifier's predictions $\hat{p}(\mathbf{x}_i)$ achieve an empirical Brier score no larger than that of the base-rate predictor, then by definition $BS = n^{-1} \sum_i (\hat{p}(\mathbf{x}_i) - y_i)^2 \leq n^{-1} \sum_i (\bar{y} - y_i)^2 = \bar{y}(1 - \bar{y})$, using the identity just proved for the right-hand side. No population expectation or law-of-large-numbers argument is needed: the comparison holds exactly, on the observed sample, by the premise that the classifier's Brier score does not exceed the base rate's.

The quantity $\bar{y}(1 - \bar{y})$ therefore plays the same role as the total variance $\sum (y_i - \bar{y})^2/n$ in regression: it is the score achieved by the trivial baseline predictor. Provided $0 < \bar{y} < 1$ (so the denominator is nonzero; if every observed response belongs to the same class, $\bar{y}(1 - \bar{y}) = 0$ and the base-rate predictor already achieves $BS = 0$, making the skill score degenerate), a Brier skill score analogous to R^2 is $BSS = 1 - BS/[\bar{y}(1 - \bar{y})]$, which equals 1 for perfect prediction ($BS = 0$) and 0 for the base-rate classifier. (Note that perfect *calibration* alone does not imply $BS = 0$: a classifier that correctly predicts $\hat{p} = 0.7$ for every observation in a group that is genuinely 70% positive is perfectly calibrated for that group but still incurs nonzero Brier loss, since $BSS = 1$ specifically requires perfect prediction, not merely correct calibration.) $\square$

Exercise 7.5. A model with three features has produced the following value function $v_{\mathbf{x}}(S)$, for every coalition $S \subseteq \{1, 2, 3\}$:

$$\begin{aligned} v(\emptyset) &= 10, & v(\{1\}) &= 14, & v(\{2\}) &= 16, & v(\{3\}) &= 13, \\ v(\{1, 2\}) &= 24, & v(\{1, 3\}) &= 19, & v(\{2, 3\}) &= 21, & v(\{1, 2, 3\}) &= 30. \end{aligned}$$

Using Definition 6.3, compute the Shapley value ϕ_i of each feature $i \in \{1, 2, 3\}$ by hand. Verify that your three values satisfy the efficiency property of Proposition 6.5.

Solution. With $d = 3$, the coalition-size weights $|S|!(d - |S| - 1)!/d!$ are $1/3$ for $|S| = 0$, $1/6$ for $|S| = 1$, and $1/3$ for $|S| = 2$.

For feature 1, the relevant coalitions $S \subseteq \{2, 3\}$ are $\emptyset, \{2\}, \{3\}, \{2, 3\}$:

$$\begin{aligned} \phi_1 &= \frac{1}{3}[v(\{1\}) - v(\emptyset)] + \frac{1}{6}[v(\{1, 2\}) - v(\{2\})] + \frac{1}{6}[v(\{1, 3\}) - v(\{3\})] + \frac{1}{3}[v(\{1, 2, 3\}) - v(\{2, 3\})] \\ &= \frac{1}{3}(4) + \frac{1}{6}(8) + \frac{1}{6}(6) + \frac{1}{3}(9) = \frac{4}{3} + \frac{4}{3} + 1 + 3 = \frac{20}{3}. \end{aligned}$$

By the same pattern,

$$\begin{aligned} \phi_2 &= \frac{1}{3}(6) + \frac{1}{6}(10) + \frac{1}{6}(8) + \frac{1}{3}(11) = 2 + \frac{5}{3} + \frac{4}{3} + \frac{11}{3} = \frac{26}{3}, \\ \phi_3 &= \frac{1}{3}(3) + \frac{1}{6}(5) + \frac{1}{6}(5) + \frac{1}{3}(6) = 1 + \frac{5}{6} + \frac{5}{6} + 2 = \frac{14}{3}. \end{aligned}$$

Efficiency check: $\phi_1 + \phi_2 + \phi_3 = \frac{20}{3} + \frac{26}{3} + \frac{14}{3} = \frac{60}{3} = 20$, and indeed $v(\{1, 2, 3\}) - v(\emptyset) = 30 - 10 = 20$, confirming Proposition 6.5. $\square$

Further Reading

Kohavi (1995) gives a classic empirical comparison of the bias-variance trade-off between k -fold and leave-one-out cross-validation and remains a standard reference for choosing k . Dietterich (1998) introduces the 5×2 cross-validation test and gives a careful comparison with McNemar’s test and resampled t -tests, including the statistical reasons the naive approaches fail. Niculescu-Mizil and Caruana (2005) is the standard empirical reference on the calibration properties of common classifiers, including the characteristic (and not always overconfident) patterns seen in bagged tree ensembles, and the choice between Platt scaling and isotonic regression. Lindholm et al. (*SML*, smlbook.org) Chapter 4 and James et al. (*ISLP*, statlearning.com) Chapter 5 both give textbook treatments of cross-validation and the train/validation/test protocol at a level consistent with this lecture. On the addendum: Shapley (1953) is the original game-theoretic source of the Shapley value and its uniqueness theorem. Lundberg and Lee (2017) unify Shapley values with several earlier feature-attribution methods under the SHAP framework and introduce KernelSHAP; Lundberg et al. (2018, 2020) introduce TreeSHAP and the consistency property discussed in Remark 6.6. Molnar’s *Interpretable Machine Learning* (the Shapley Values chapter of the continuously updated online edition, cited below by its most recent print edition) gives an accessible, code-oriented treatment of Shapley values alongside other interpretability methods, and is the recommended first stop for students wanting worked examples beyond this lecture.

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Molnar, C., 2025. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd edn. <https://christophm.github.io/interpretable-ml-book>